Residential Thermal Prediction with Limited and Noisy Sensor Data

Yingzi Xu | CEE 292: Scientific Machine Learning | Individual report | 6 November 2026

Code, data and results: https://github.com/yingzixu233/residential-thermal-benchmark

1. Research question and contribution

Can a small neural model predict residential temperatures with only a few days of data, and do tree ensembles or noise augmentation help? I implement the direct MLP transition in Vasisht et al. [1], Section 3.2, Eq. (6), and extend it with Random Forest (RF), XGBoost, a temperature-noise ablation and Shapley explanations. This method-level adaptation follows the outline's public residential case [2,3]. The original commercial-emulator data are available on request; its full benchmark, latent-state and physics-constrained architectures, and multi-step curriculum are outside this implementation.

2. Data and evaluation

The unchanged house_data.csv has 385 hourly observations from 30 March to 15 April 2019, with no missing values. Twenty inputs comprise nine zone temperatures (C), nine heating energies (Wh per hourly interval), outdoor temperature (C) and solar irradiance (W/m2). Targets are the nine next-hour temperatures; row-t heating/weather are aligned to t+1. The public archive checksum and CSV SHA-256 are recorded.

Training observations are [0,240), validation [240,288), and test [288,385), using zero-based, stop-exclusive indices. Recent 3/5/10-day windows yield 71/119/239 training pairs; validation and test have 47 and 96. Split-boundary transitions are excluded. Scalers fit training pairs only. Hyperparameters and checkpoints minimize clean one-step validation RMSE, without validation refitting or test-based tuning.

Seeds 0,1,2 and Gaussian temperature-input noise of SD 0/0.2/0.5 C give 162 model-budget-seed-noise settings. Targets remain recorded measurements; noise is paired across models and budgets. One-step prediction refreshes measured inputs each hour. Recursive 24-hour forecasts use 73 complete, overlapping test origins, observe temperature only at initialization, and then feed back predictions. Noise is added only at initialization during rollout. Recorded future heating/weather are supplied, so forecasts are conditional on known forcing.

3. Models and training

Direct transition: $T(t+1) = f_\theta([T(t), Q(t), T_{out}(t), GHI(t)])$.

The PyTorch MLP is 20-32-32-9 with Tanh hidden layers, a linear output, and training-only input/target standardization. It predicts absolute temperature. Full-batch Adam uses learning rate 0.003 or 0.01, weight decay 10^-4, at most 1500 epochs, checks every 10 epochs and 200-epoch patience. MLP + noise adds fresh 0.2 C Gaussian noise to training temperature inputs each epoch. The 32-unit layers follow the outline; the original paper searches multiple architectures.

Baselines are persistence and standardized Ridge (alpha 0.1/1/10/100). RF [5] has 200 multi-output trees; (depth, leaf size) is (6,2) or (unlimited,3). XGBoost [4] fits nine scalar outputs using all inputs: 250 trees, depth 2 or 3, learning rate 0.03, lambda 5, subsample 0.9. These small searches and the one-step objective are project choices, not exhaustive or equal optimization budgets.

4. Results overview

Table 1. Ten-day training, clean inputs: mean (sample SD) over three seeds, in C. Path metrics pool 73 origins x 24 horizons x 9 zones; h=24 uses the terminal hour. One-step metrics use 96 transitions. Seed SD is not a confidence interval.

Model	1 h RMSE	24 h path MAE	24 h path RMSE	h=24 RMSE
Persistence	0.654 (0.000)	1.392 (0.000)	1.760 (0.000)	1.761 (0.000)
Ridge	0.258 (0.000)	0.307 (0.000)	0.401 (0.000)	0.415 (0.000)
MLP	0.409 (0.037)	0.480 (0.041)	0.627 (0.058)	0.633 (0.060)
MLP + noise	0.474 (0.016)	0.482 (0.017)	0.630 (0.024)	0.665 (0.019)
RF	0.638 (0.027)	0.727 (0.017)	1.011 (0.023)	1.058 (0.027)
XGBoost	0.533 (0.003)	0.723 (0.019)	1.011 (0.022)	1.151 (0.035)

Key finding. As shown in Table 1, Ridge achieves the lowest error across all four reported metrics, followed by the standard MLP. Greater model complexity therefore does not improve clean-input performance in this experiment.

4. Results: data budget, forecast horizon and sensor noise

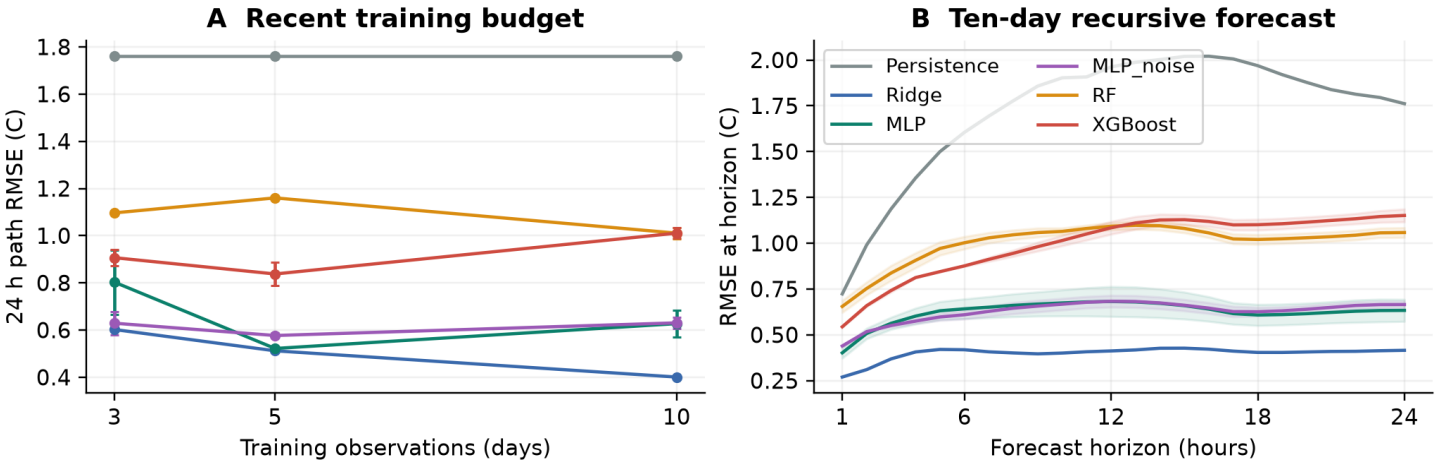


Figure 1. Clean-input path RMSE versus recent training budget (left), and RMSE at each recursive horizon after ten-day training (right). Bars/shading show sample SD across three seeds. Horizon 1 uses 73 origins, so it differs from Table 1's 96-point one-step metric.

Complexity and long forecasts. As shown in Table 1 and Figure 1 (right), Ridge has the lowest clean ten-day path RMSE (0.401 C), versus 0.627 C for MLP and about 1.011 C for both tree ensembles. This is 77.2% below persistence (1.760 C). XGBoost beats RF at one step, but their path errors are nearly identical. One-step rankings therefore do not determine recursive accuracy; Figure 1 (right) supports retaining a regularized linear baseline for long forecasts.

Limited data. Figure 1 (left) shows that, at three days, augmentation reduces MLP path RMSE from 0.803 to 0.629 C (21.7%). At five days, standard and augmented MLP reach 0.522 and 0.577 C; at ten days, 0.627 and 0.630 C. Thus, the benefit of augmentation is regime-dependent (see Figure 1, left). Longer history is not uniformly better because the recent-data windows differ in both sample count and operating conditions.

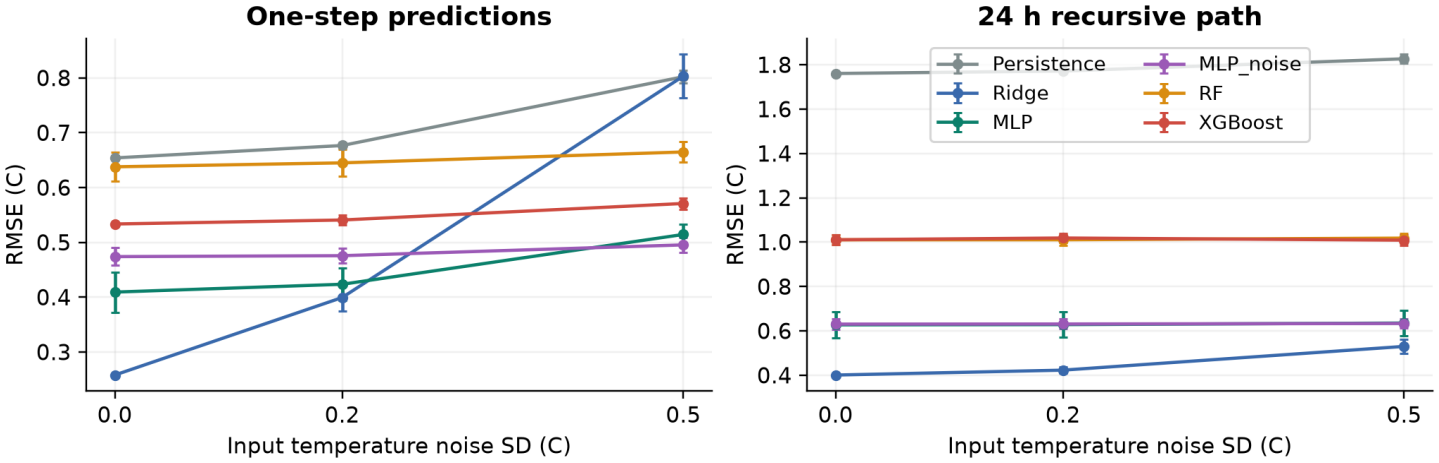


Figure 2. Ten-day training under added indoor-temperature noise. One-step forecasts refresh perturbed measured states hourly (left); recursive forecasts perturb only the initial state (right). All model families use paired perturbations. Error bars show sample SD over three seeds.

Noise robustness. As shown in Figure 2 (left), at 0.5 C input noise, one-step RMSE is 0.495 C for augmented MLP, 0.514 C for standard MLP, and 0.803 C for Ridge. Augmentation sacrifices some clean-input accuracy while reducing sensitivity to measurement perturbations. Ridge's recursive path RMSE is still 0.530 C (see Figure 2, right). Since noise is not re-injected inside rollouts, the panels address different deployment conditions. No model uses physical clipping or a stability constraint.

Summary. Table 1 and Figures 1-2 show that model selection depends on data availability and sensor quality: Ridge is strongest with clean observations, while augmented MLP is more tolerant of noisy one-step inputs.

5. Qualitative forecast trajectories

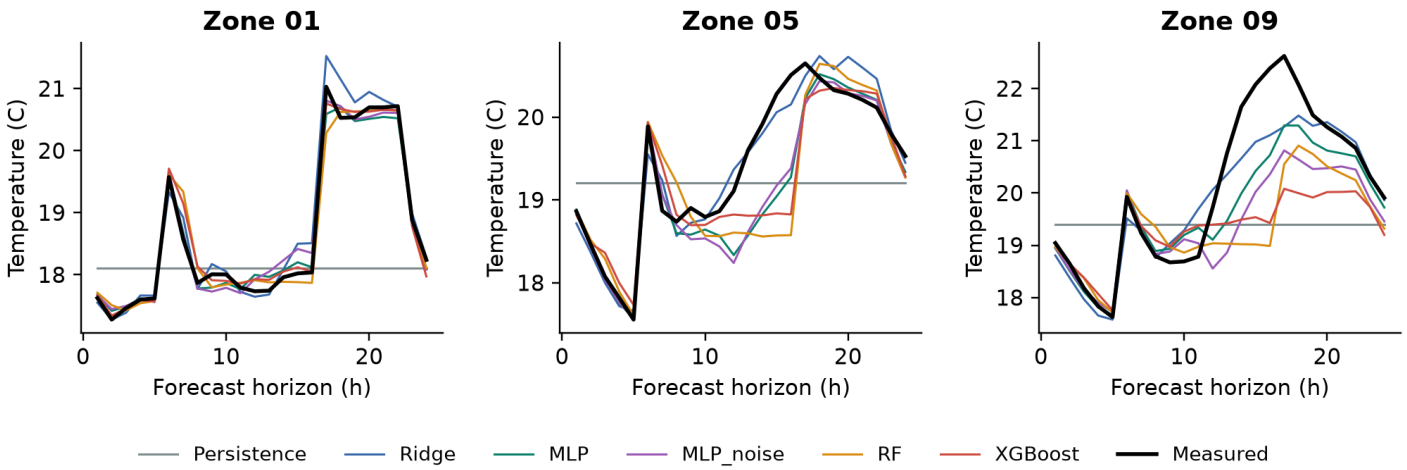


Figure 3. First test origin (11 April 2019, 00:00), ten-day training, seed 0, clean inputs: 24-hour predictions for zones 01, 05 and 09. Black curves are measured temperatures. The fixed example illustrates trajectory shape; aggregate results use all 73 origins and all nine zones.

As shown in Figure 3, models follow many of Zone 01's abrupt changes but all learned models underestimate the observed Zone 09 peak in this example. This highlights local errors hidden by aggregate scores in Table 1. The fixed example does not establish performance at every origin or zone.

6. Shapley interpretation

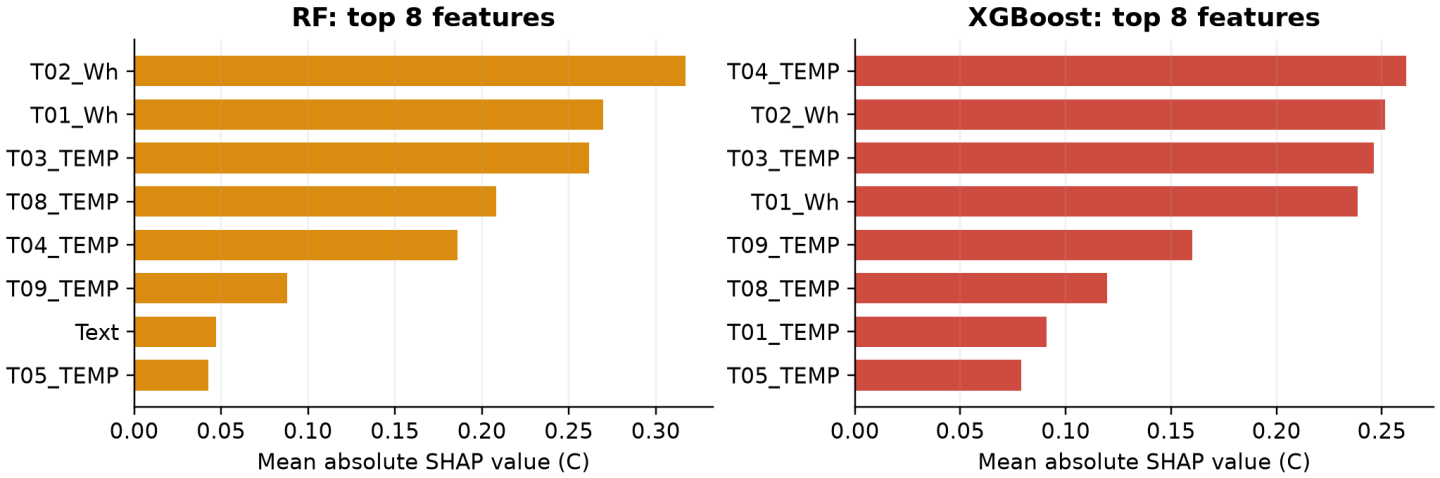


Figure 4. Top eight mean absolute TreeSHAP contributions for RF and XGBoost. Values are averaged over 96 clean test inputs and nine outputs for the ten-day, seed-zero models, with a common background of 100 evenly spaced training inputs.

Interventional TreeSHAP [6] decomposes one-step absolute temperature predictions into a background expectation and feature contributions (C). All 864 predictions/model pass reconstruction within 10^-4 C. As shown in Figure 4, RF's largest attribution is T02 heating (0.317 C), whereas XGBoost's is T04 temperature (0.261 C). Temperature features account for 54.2%/61.9% of total mean absolute attribution, and heating for 42.0%/32.7% (RF/XGBoost). Figure 4 therefore highlights different learned dependencies; these attribution shares are not energy shares or explained variance.

Correlated temperatures and thermostat feedback prevent causal interpretation: cross-zone importance does not establish thermal adjacency. Interventional masking can create unrealistic feature combinations, and rankings depend on the fitted seed and background. The explanations describe one-step outputs, not full recursive trajectories.

7. Limitations and conclusion

One house, sixteen days, short validation and overlapping rollouts limit generalization. Three seeds capture algorithm/noise variability rather than building or seasonal uncertainty. Known future forcing, noisy labels and small search grids further limit the conclusions. Overall, Table 1 and Figures 1-2 favor Ridge for clean inputs and targeted MLP augmentation under scarce or noisy data. Figures 3-4 add trajectory-level checks and model interpretation. Broader claims require additional buildings, blocked temporal evaluations and uncertain-forcing tests.

References

Supporting references for the three-page main report

[1] S. Vasisht et al. Physics vs structure: A systematic benchmark of learning strategies for multi-zone building thermal dynamics. Energy and Buildings 361, 117472 (2026). doi:10.1016/j.enbuild.2026.117472.

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